



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract:

Online assessment has become an important part of modern education, providing students with a convenient way to participate in quizzes and allowing institutions to evaluate learning outcomes efficiently. However, traditional quiz management methods and basic online quiz applications often face challenges related to secure user authentication, efficient question management, result processing, scalability, and maintaining different functionalities within a single application. These limitations highlight the need for a reliable, secure, and scalable online quiz management system that can efficiently handle the complete quiz lifecycle.

To address these challenges, the proposed Online Quiz System is designed as a web-based application using a microservices architecture. The system provides separate services for managing users, quizzes, questions, and results, allowing each functionality to operate independently. The application is developed using Spring Boot and Spring Cloud, with Eureka Server used for service discovery and Spring Cloud Gateway used for centralized API routing. This architecture improves modularity, maintainability, scalability, and communication between different services.

Security is an important aspect of the proposed system. JWT-based authentication and role-based authorization are implemented to ensure that only authorized users can access specific functionalities. Students can securely access available quizzes, submit answers, and view their results, while administrators can manage quizzes, questions, users, and other system information. MySQL is used for database management, following a database-per-service approach to maintain service independence.

The system also incorporates modern software development and testing tools to improve development quality and reliability. Postman is used for REST API testing, while Git and GitHub support version control and collaboration. GitHub Actions is used to implement CI/CD practices, enabling automated build and testing workflows. Jira is used for user stories and sprint planning, Lucidchart AI is used for designing the system architecture, GitHub Copilot supports AI-assisted development and unit-test generation, and Semgrep is used for security analysis.

Overall, the proposed Online Quiz System provides a secure, scalable, and maintainable platform for conducting online assessments. By combining microservices, REST APIs, authentication, database management, automated testing, security analysis, and CI/CD practices, the project demonstrates the application of modern software engineering principles for developing a reliable online assessment platform.

Project Architecture/Project Overview:

